

Bridge Damage Detection in Presence of Varying Temperature Using Two-Step Neural Network Approach

Smriti Sharma¹ and Subhamoy Sen²

Abstract: The dynamic properties of bridges can be affected not only through damage but also from ambient uncertainty. False-positive or negative alarms may be raised if environmental effects are not considered in the detection algorithm. This article presents a two-step data-driven approach that can incorporate temperature effects in vibration-based damage detection and localization, provided the temperature is also measured. To detect the occurrence of damage, prediction errors of an autoassociative neural network (AANN) framework are first employed as a temperature-invariant novelty index (NI). Further, for damage localization, NIs associated with each of the damage cases, are classified using a radial basis function neural network (RBFNN). The proposed algorithm is numerically tested on a multispan bridge structure involving measurement uncertainties. The performance of the RBFNN classifier is further assessed using different classifier performance metrics. The possibility of positive and negative false alarms raised by the proposed algorithm and its sensitivity to measurement noise contamination is also investigated. It is observed that the proposed data-based approach can efficiently isolate a fault, even in the presence of temperature variation. DOI: [10.1061/\(ASCE\)BE.1943-5592.0001708](https://doi.org/10.1061/(ASCE)BE.1943-5592.0001708). © 2021 American Society of Civil Engineers.

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Introduction

Structural damage may induce changes in the dynamic properties of a structure; therefore, the occurrence and location of damage can be sensed by analyzing its dynamic responses. During the normal operation of a structure, damage-sensitive features, such as modal information, are extracted and archived to standardize its healthy state (Chakraborty and DeWolf 2006; Cardini and DeWolf 2009). This health information can later be compared with corresponding information collected when the structure is perceived to be damaged for further confirmation and subsequent damage localization (Weinstein et al. 2018).

For structural health monitoring (SHM), modal domain information is traditionally preferred over response time history, since the modal parameters are easy to store and work with and can be used to develop prompt damage detection approaches (Sohn et al. 2004). Nevertheless, the impact of damage on modal properties is not always prominent or at least monotonic. Being global parameters, the modal properties typically have mild sensitivity toward local structural changes. It is, therefore, difficult to assess structural health based only on modal properties (Salawu 1997).

Moreover, not only damage, but environmental agents, such as temperature, wind, humidity, and even service conditions (e.g., traffic) can also affect the dynamic properties of a structure (Alampalli 2000; Farrar and Jauregui 1998). Among the different infrastructures, bridges are significantly exposed to extreme environmental conditions. The employment of any SHM approach

should thus be robust against environmental uncertainties. Isolation of damage-induced changes in modal properties from the overall variations due to all aspects combined, however, poses a major challenge in SHM (Kullaa 2009).

In this study, among several environmental agents, the effect of ambient temperature on damage detection in a bridge structure is investigated. For damage detection, a model-based SHM approach is employed, in which a sufficiently accurate predictor model simulates the damaged response. Since the performance of the employed SHM approach depends on the quality of this supporting predictor model, the dependence of structural stiffness (material as well as geometric) on temperature should be modeled precisely. Typically, material properties change with temperature, which, in turn, affects the material stiffness. The geometric stiffness, however, arises as a result of the temperature-induced internal stresses in the structure and, thereby, its dependence on temperature is very much case-specific.

As an example, for a fixed-fixed beam, lateral restraints restrict thermal elongation, resulting in an axial prestress in the beam that, in turn, alters its geometric stiffness. For the same beam with one end support axially released, prestress will not develop, causing no change in geometric stiffness. Again, with nonuniform temperature over the beam cross section, the thermal gradient induces a curvature, owing to the differential in the induced prestress. For a fixed-fixed boundary condition, temperature-induced curvature close to the supports becomes nullified by the end moments, affecting the geometric stiffness. While for a beam with laterally free but rotationally restricted end supports, geometric stiffness will be affected because of the compensatory end moments, with laterally restricted but rotationally free end supports, geometric stiffness will remain unaffected. This case specificity condition of the temperature effects on geometric as well as material stiffness makes the problem challenging as well as interesting.

Regardless of the case, the variation in temperature, however, always affects the overall structural stiffness through material or geometric stiffness, leading to variation in dynamic properties even when the structure is undamaged. In usual practice, the effect

¹i4S Laboratory, Indian Institute of Technology Mandi, Kamand, HP 175005, India.

²i4S Laboratory, Indian Institute of Technology Mandi, Kamand, HP 175005, India (corresponding author). ORCID: <https://orcid.org/0000-0001-8021-6693>. Email: subhamoy@iitmandi.ac.in

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of temperature on structural stiffness is modeled only partially (through material stiffness) in most studies. The geometric component of structural stiffness is mostly ignored during the modeling. To enhance the reality replication prospective of the supporting model, both the stiffness components along with their temperature dependency should ideally be taken into consideration.

In this study, an SHM approach based on machine learning (ML) is proposed to detect and localize damage in a multispan bridge in the presence of varying temperatures using modal domain information. The associated difficulty with modal information has already been discussed, as it further intensifies with varying temperature (Xu et al. 2010). ML helps in this regard through the extraction of damage-sensitive yet temperature-invariant features embedded in the modal information to sense and isolate damage.

With this objective, the current study employs a two-step data-based approach: the first step employs a prediction framework based on an autoassociative neural network (AANN) to detect the occurrence of damage, if any, while the second step uses prediction error patterns for damage localization with a radial basis function based neural network (RBFNN). The AANN is trained to yield a relatively large prediction error norm as a novelty index (NI) in the presence of damage. NIs associated with each of the damage cases are further classified using a RBFNN trained a priori using a rich archive of training data corresponding to the damaged and healthy states of the structure.

The following section discusses the ML-based SHM approaches in the existing literature. Then follows a discussion of the impact of temperature on structural stiffness. Next, the proposed approach is discussed in detail, prior to a demonstration of the numerical experimentation. Finally, some conclusions are presented.

Literature Review

With the advent of cheap and efficient computation technologies, artificial intelligence has emerged as an alternative to phenomenological model-based techniques for damage detection. The relative benefit of ML over traditional model-based SHM is that, unlike the latter, the former approach does not demand costly model simulations online, making the detection process prompt as well as robust. Modern SHM techniques are, therefore, employing data-based approaches in abundance to find a solution for real-life SHM problems. Of the ML-based approaches, artificial neural networks (ANNs), which are basically a numeric imitation of functional aspects of the human nervous system, have been widely used over the past few decades for SHM applications (Barai and Pandey 1995; Mehrjoo et al. 2008). The ANN trains a network using knowledge embedded in a measured dataset to predict for future situations. This model-free learning-based prediction makes the use of ANNs a potential approach for real-life SHM of complex structural systems. A sophisticated and well-trained ANN can model any complex and weakly understood interrelationship between structural health and measured responses for which the modeler has limited or no information (Cao et al. 1998). Several parametric and nonparametric ANN-based SHM techniques have been developed in the past few decades (Zhou et al. 2011; Figueiredo et al. 2011; Kullaa 2009; Mehrjoo et al. 2008). Parametric ANN-based approaches have proved efficient in isolating damage-sensitive patterns in structural responses and further classifying them to specific damage cases (Jin et al. 2016; Santos et al. 2016).

Over the last few decades, ANNs have been used as predictors or classifiers in SHM research to either predict or classify a chosen damage-sensitive feature (Weinstein et al. 2018). The use of radial basis functions (RBFs) provides an effective alternative to

conventional ANN-based SHM to classify unique damage-sensitive patterns in the inputs to corresponding damage cases. RBF networks are universal approximators and have the capability to train very quickly. Okafor and Dutta (2001) proposed an RBFNN to estimate the size and depth of damage in plates. Reddy and Ganguli (2003) proposed an RBFNN for modal frequency-based damage detection in helicopter rotor blades. Khan and Parhi (2015) used similar approaches for fault detection in a composite beam from measured modal properties. Zang et al. (2007) proposed an integrated method for damage assessment based on correlation functions and an RBF network. Machavaram and Shankar (2013) employed a two-stage improved RBF (IRBF) for substructural joint damage detection in both frequency and time domains.

The impact of temperature on structural health, especially for bridges, has drawn significant interest in recent research (Glisic et al. 2011; Kromanis et al. 2016; Yarnold and Moon 2015; Abdel-Jaber and Glisic 2019; Zhou and Sun 2019). Zhou and Sun (2019) investigated a cable-stayed bridge to monitor the effect of temperature on cables, girders, and towers of the bridge, employing forward models involving temperature-dependent material models and nonlinearity. In their study, the case specificity of temperature effects is elaborately manifested. Kromanis et al. (2016) conducted similar research on developing a physics-based model to understand the effect of temperature on the bearings of the Cled-dau Bridge. Yarnold and Moon (2015) used a simplified numerical model as a baseline to understand the correlation between temperature variation and structural responses (strain and displacement). Provided the vicinity of the damage is measured, their proposed method can detect its location efficiently with the help of the baseline model. This study made a comparison between traditional vibration-based SHM and temperature-based SHM in order to justify the need for taking temperature into consideration. In continuation with their earlier research (Glisic et al. 2011), Abdel-Jaber and Glisic (2019) proposed a method for on-site assessment of pre-stress loss in the Streicker Bridge at Princeton University using fiber-optic sensors and 7-years monitored data. Overall, the majority of these methods can be categorized as forward approaches that investigate the real structure or its model to develop an understanding of the effect of ambient temperature on structural performance.

Damage detection in the presence of temperature variation has also been studied using data-based inverse approaches (Zhu et al. 2019; Gu et al. 2017; Ozdagli and Koutsoukos 2019; Moaveni and Behmanesh 2012; Alampalli 2000). Zhu et al. (2019) proposed a temperature-driven anomaly detection technique using temperature-dependent moving principal component analysis to separate temperature influences and detect the damage in a structure. Attempts have also been made to develop a predictor network trained to mimic the functional relationship between temperature variation and modal properties (Jin et al. 2016) under normal operational conditions. Later, an NI was defined to measure the departure of the measured parameter from the prediction. A significant change in the NI was subsequently attributed to a possible indication of damage occurring in the system, thereby eliminating the effect of ambient uncertainty from the detection process. Taking this as a basis, Ozdagli and Koutsoukos (2019) proposed a novel measure to detect damage under temperature uncertainty. Gu et al. (2017) proposed an AANN-based approach for detecting damage in the presence of varying temperature.

Clearly, the mentioned SHM methods emphasize either a sensitivity-based approach involving a detailed baseline model for precise detection or a novelty identification in a measured response for prompt detection of damage occurrence. Further, the ML-based approaches discussed here (Zhu et al. 2019; Gu et al. 2017; Ozdagli and Koutsoukos 2019; Moaveni and Behmanesh 2012; Alampalli 2000) can perfectly detect an occurrence of

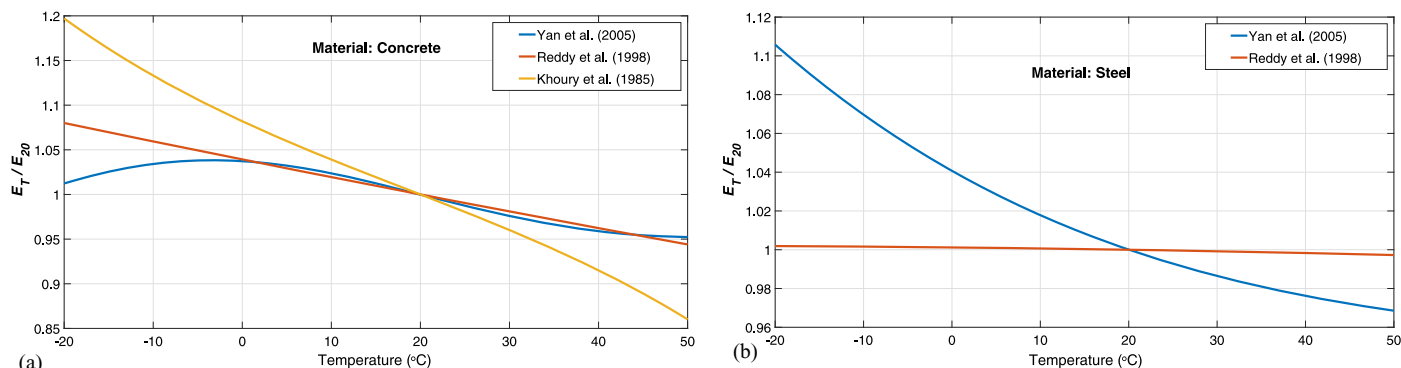


Fig. 1. Effect of temperature on elastic modulus: (a) concrete; and (b) steel.

damage but, as such, cannot localize it. However, a practical solution to SHM problems demands the location of damage to be additionally identified. This study, therefore, proposes a data-based algorithm coupling an AANN with an RBFNN to exploit their relative potential in achieving temperature robustness and to classify damage precisely.

Nevertheless, for a data-based approach, the availability of a rich archive of damaged as well as undamaged responses is a precondition for accuracy. While the undamaged dataset can be collected from prolonged monitoring of the structure, the damaged dataset is not always available in abundance. This can be complemented with a good phenomenological model, which takes the temperature-dependence of structural stiffness into consideration, to synthetically generate damaged data under different damage scenarios and temperature conditions. A good predictor model also helps in understanding the intricate functioning and interrelationship of all the contributing aspects and their variations. The precision of a data-based SHM approach trained with a synthetic damaged response evidently depends on the quality of the predictor model. It is, therefore, advisable to calibrate or update the predictor model using measurements collected from the real structure prior to its use. The following section details the temperature-dependence of structural stiffness embedded within the employed predictor model.

Temperature Effect on Dynamic Properties of Structures

Stiffness strongly influences the dynamic properties of a structure. Also, its dependence on temperature is more significant than mass or damping. Temperature may affect the stiffness in several ways: material, geometry, and boundary conditions are a few of them. Temperature-induced changes in elastic modulus, Poisson's ratio, and density may cause changes in material stiffness and, subsequently, modal properties, although not in a linear manner. Several experimental investigations have been conducted to explore the dependency of elasticity of concrete and steel on temperature (Reddy and Chin 1998; Yan et al. 2005). Results for a few of these are presented in Fig. 1.

For beam-like structures, such as bridges, a significant temperature change can cause supports to move, owing to the bridge's axial expansion or shortening. Temperature variation may also induce changes in boundary conditions or structural configuration, leading to prestress in the structure if the movement is restrained (Alampalli 2000). Prestress leads to nonlinear behavior in the structure, affecting its geometric stiffness. The structure might also experience a thermal gradient over its cross section, leading to a curvature in the structure (also known as thermal bowing) if the rotation in the supports is not restrained. Otherwise, a tensile force

will be developed to correct the curvature. To investigate the effects of temperature on the geometric stiffness of a beam-like structure (e.g., a bridge), two types of heating regime are considered:

- uniform temperature rise: $\Delta T = T_{\text{avg}} - T_0$ with $T_{\text{avg}} = (T_2 + T_1)/2$
- linear thermal gradient along depth d : $T_y = |(T_2 - T_1)|/d$

where T_1 = temperatures of the top surface of the bridge; T_2 = temperature of the bottom surface of the bridge (usual phenomena due to differential solar heating of deck slab and soffit); T_{avg} = mean temperature; T_0 = reference temperature; and T_y = the gradient of this assumed linear temperature variation. Only the temperature gradient along the vertical direction is considered; no gradient is assumed along the transverse direction.

Usually, bridge supports are idealized as fixed (all degrees of freedom restrained), hinged (with rotational degrees of freedom released), or roller (with axial degrees of freedom and rotational degrees of freedom released). Yet these idealizations seldom conform to reality. For the sake of practicality, this article assumes a bridge [Fig. 2(a)] with a fixed support at one end, while its other support is finitely restrained in axial degrees of freedom and fully restrained in rotational degrees of freedom. The finite restraint is modeled using a linear spring of stiffness k_t at the support end. For such a support condition, the axial prestress σ_a [Fig. 2(b)] can be obtained as

$$\sigma_a = \frac{E\alpha\Delta T}{1 + EA/k_t l} \quad (1)$$

where l = length of the bridge; A = cross section of the bridge; α = coefficient of thermal elongation; and E = modulus of elasticity of the material. Similarly, the bowing induced strain ϵ_ϕ and associated tensile stress σ_b can be modeled [Fig. 2(c)] as

$$\epsilon_\phi = 1 - \frac{\sin(l\phi/2)}{l\phi/2}; \quad \sigma_b = E\epsilon_\phi \quad (2)$$

where ϕ = radius of bowing induced curvature, which can be obtained as $\phi = \alpha T_y$. Evidently, for a positive change in temperature, the resistive stresses induced due to the thermal expansion and bowing are opposite to each other, resulting in an effective stress σ_t as the difference between the two opposing stresses, i.e., $\sigma_t = \sigma_b - \sigma_a$. The resulting prestress should be incorporated in the predictor model through its impact on the geometric stiffness matrix (Gavin 2011).

Proposed Approach

In this study, two separate networks (AANN and RBFNN) are employed in sequence: the first to detect the presence of damage and the second to localize it. The overall idea is to first train an AANN-

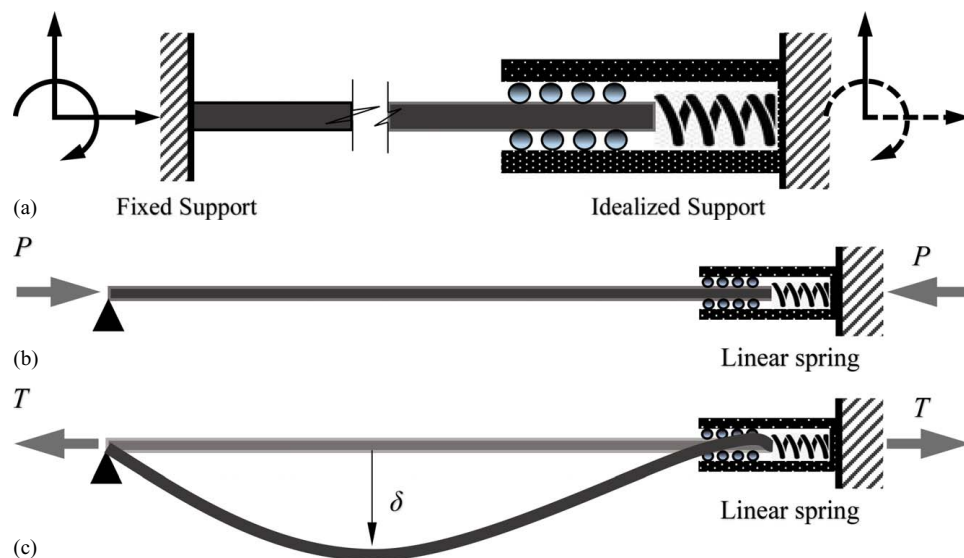


Fig. 2. Idealized support and effects of temperature: (a) idealized bridge support with finite linear restraint; (b) temperature-induced prestress; and (c) bowing effect due to thermal gradient.

based predictor model using frequencies collected from a healthy structure under varying temperature and then to use this network with frequencies collected from a structure in an unknown state to obtain the prediction error, in order to (i) define an NI for detection of damage occurrence, and (ii) localize any damage with an RBFNN classifier.

Autoassociative Neural Network (AANN)

Recent trends in data-based SHM research involving environmental uncertainty are concerned more about the tracking of certain novelty measures (Worden 1997; Zhou et al. 2011) to detect anomalies in structures. A recursive comparison of certain patterns is performed between new and reference healthy data in order to identify a significant change, termed a “novelty.” The use of AANNs can be identified as an efficient approach for novelty detection, in which a network is trained to predict the output exactly the same as the input. Apart from input and output layers, a typical AANN consists of three hidden layers: a demapping layer with more nodes than the input dimension, a bottleneck layer with a dimension smaller than the input layer, and a mapping layer of an equal number of nodes as in the demapping layer.

A typical AANN has symmetric architecture on both sides of the bottleneck layer. The information in the input layer becomes deconstructed in the higher-dimensional demapping layer and subsequently compressed in the bottleneck layer. The same is repeated but in a reverse way to get back the inputs. Thereby, the network learns the interrelation between inputs that originate from similar sources. The prediction error thus remains small as long as the assumption on the source similarity holds true. Any alterations in the source will eventually cause a larger error in the prediction, signifying a change in the input source.

AANN-Based Damage Detection

This study employs a network similar to the conventional AANN with an asymmetric modification by augmenting ambient temperature in the input layer in addition to the healthy natural frequencies. A typical five-layer (input, output, and three hidden layers) feedforward back-propagation network is constructed [Fig. 3(a)] with

tangent-sigmoid transfer functions for the hidden nodes and a Levenberg–Marquardt algorithm as the training function (Kostić and Gül 2017) to learn the temperature–frequency interrelation. The network is trained using only healthy frequencies. Violation of the healthy state will eventually invalidate this prediction model, causing a large prediction error ϵ_k for frequencies corresponding to damaged structures. The norm of ϵ_k can therefore be considered an NI, as an indication of damage in the system. For the k th measured frequency set, $f_{k|t_k}^i$, as input, under ambient temperature t_k , and corresponding AANN predicted output $f_{k|t_k}^o$ as $f_k^o = \text{AANN}(f_k^i, t_k)$ (superscripts i and o denote the input and output, respectively), the NI for damage detection can be defined as

$$\text{NI} = \|\epsilon_k\|; \quad \text{where } \epsilon_k = (f_{k|t_k}^i - f_{k|t_k}^o)(f_{k|t_k}^i)^{-1} \quad (3)$$

The AANN is trained using a rich archive of healthy frequencies and the maximum NI recorded during the training phase is selected as the limiting NI for any set of frequencies to be categorized as undamaged or healthy. Through numerical experimentation (discussed later), it is identified that, unlike the impact of local damage on modal frequencies, the impact on the AANN prediction error is prominent and consistent, even under varying temperature. Every unique damage scenario is observed to demonstrate a unique pattern in the AANN prediction error, even when the change in modal frequencies is small. An RBFNN classifier is, therefore, employed to classify the error patterns to corresponding damage cases, as discussed next.

RBFNN-Based Damage Localization

Typically, an RBFNN consists of three layers with one layer hidden [Fig. 3(b)]. There are feedforward connections between the nodes in the layers. With the first and third layers being input and output layers, the hidden layer consists of radial centers obtained by clustering the inputs. With the RBFNN, the transfer functions of individual nodes are radially symmetric. The RBFNN generates the same output for all the inputs lying at the same radial distance from its radial centers; this mapping is nonlinear. However, the output layer is a linear combination of the hidden layers and therefore is linear. Compared with the traditional back-propagation neural network (BPNN), the RBFNN is known to

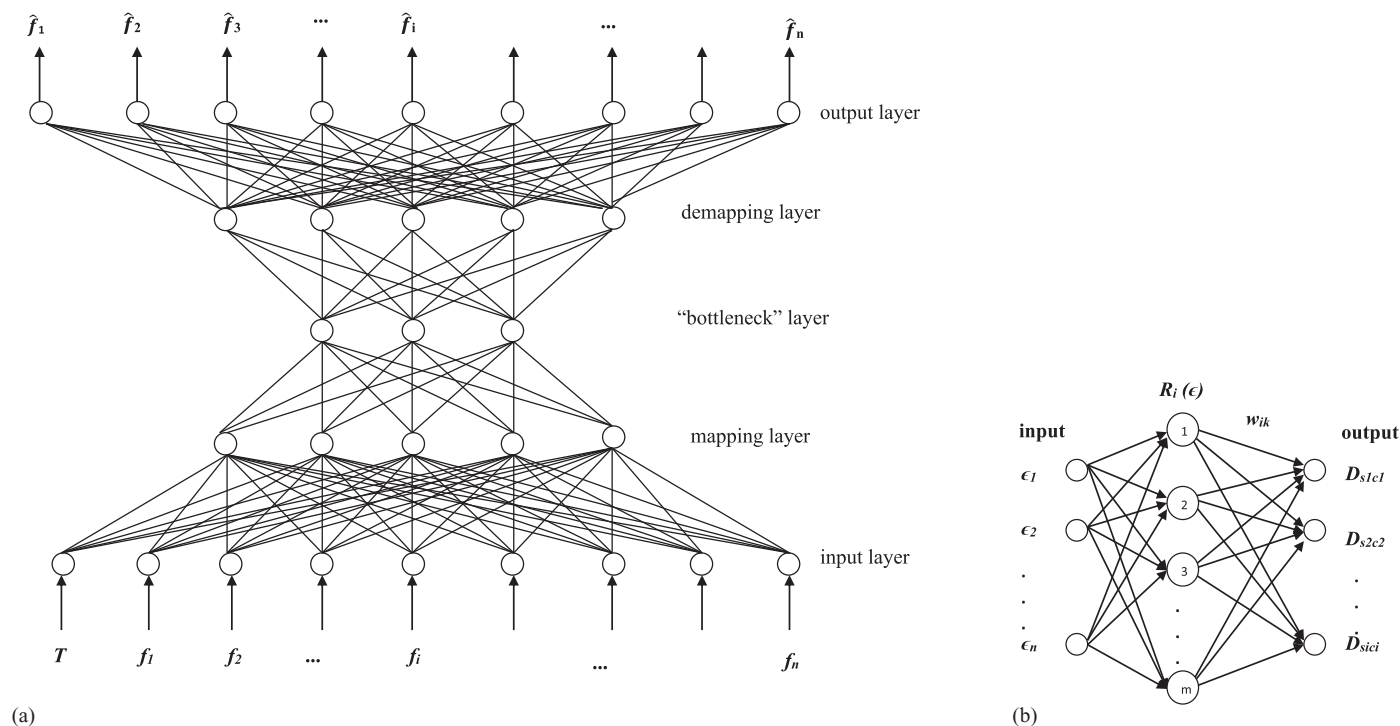


Fig. 3. Detection and localization network: (a) AANN model; and (b) RBFNN model.

be a fast-learning algorithm (Jayawardena et al. 1997; Xie et al. 2011; Fath et al. 2020). Additionally, it is characteristically a local approach mimicking local adjustment of the human nervous system.

In the input layer, unsupervised learning (k -means clustering) is performed at each input node to classify the input data among k different clusters $\{C_1, C_2, \dots, C_k\}$ based on their radial distances. The clusters are defined using their mean positions $\{c_1, c_2, \dots, c_k\}$ and variances $\{s_1, s_2, \dots, s_k\}$. The centers of the hidden layer nodes obtained from k -means clustering are the final centers, which act as the basis of the RBF. There is a family of RBF activation functions to choose from. The specific kernels that are employed for this study are detailed in the following:

$$\begin{aligned} \text{Gaussian radial function: } & \phi(\mathbf{z}_i) = e^{-(1/2)\mathbf{z}_i^T \mathbf{s}_i^{-1} \mathbf{z}_i} \\ \text{thin plate spline: } & \phi(\mathbf{z}_i) = \mathbf{z}_i \mathbf{z}_i^T \log(\mathbf{z}_i) \\ \text{quadratic function: } & \phi(\mathbf{z}_i) = (\mathbf{z}_i \mathbf{z}_i^T + 1)^{1/2} \\ \text{inverse quadratic function: } & \phi(\mathbf{z}_i) = (\mathbf{z}_i \mathbf{z}_i^T + 1)^{-1} \end{aligned} \quad (4)$$

where $\mathbf{z}_i = |\mathbf{x} - \mathbf{c}_i|$, where $\mathbf{x}$ = the input and $\mathbf{c}_i$ = the center of the i th cluster, with $i = 1, 2, \dots, k$. The output layer, being a linear combination of the hidden layers, can be presented as

$$y_j = \sum_{i=1}^k w_{ij} \phi_i(\mathbf{z}_i); \quad j = 1, 2, \dots, p \quad (5)$$

with weight w_{ij} correlating the i th hidden layer to the j th output layer; p is the dimension of the output layer.

Focusing on location-based damage identification, the entire structure is discretized into certain numbers of segments. The health of each segment is marked using Boolean values (1 = damaged; 0 = healthy). The RBFNN is then employed to classify the AANN prediction error to the corresponding damage case. Furthermore, four different RBF kernels are compared [Eq. (4)] to develop a multi-kernel RBFNN in which each individual kernel is weighted and

combined. For this, the RBFNN is first employed on a training dataset with each of the candidate kernels individually; their performances are quantified as

$$P_k = \frac{1}{n_c n_s} \sum_{i=1}^{n_c} \sum_{j=1}^{n_s} \mathbb{I}_k \{\hat{D}_{ij}; D_{ij}\} \quad (6)$$

where $\hat{D}_{ij}$ = estimated damage state defined with Boolean measure; D_{ij} = actual damage state defined with Boolean measure; n_c = number of training samples; n_s = number of segments the structure is divided into; and $\mathbb{I}_k \{a; b\}$ is a truth function that takes the value 1 if its arguments are the same and 0 otherwise. Thus, the performance measure of the k th kernel, P_k , takes a value within the range $\{0, 1\}$, with 1 denoting perfect prediction. The multi-kernel RBFNN then estimates the radial distance $\Phi(\mathbf{z})$ as

$$\Phi(\mathbf{z}) = \sum_{k=1}^4 \delta_k \phi_k^i(\mathbf{z}) \quad (7)$$

with δ_k being weights of the k th kernel $\phi_k^i(\mathbf{z})$, which can be obtained as $\delta_k = P_k / \sum_{i=1}^4 P_i$. The overall algorithm is detailed in Fig. 4.

Numerical Experimentation

The proposed methodology was tested on a numerical three-span concrete girder bridge [Fig. 5(a)] with the cross section presented in Fig. 5(c). The bridge was subjected to varying temperatures, over the range -30° to 60°C . In addition, a 3°C temperature gradient along the depth was also assumed to cause bowing. The bridge was considered to be fixed at one end and roller supported over the middle piers. The other end of the bridge was supported as defined in Fig. 2(a). The stiffness of the linear spring was considered to be 20% of the axial stiffness of the bridge girder.

The bridge was modeled using a finite-element modeling (FEM) technique as a 3D beam with the cross section presented in Fig. 5(b)

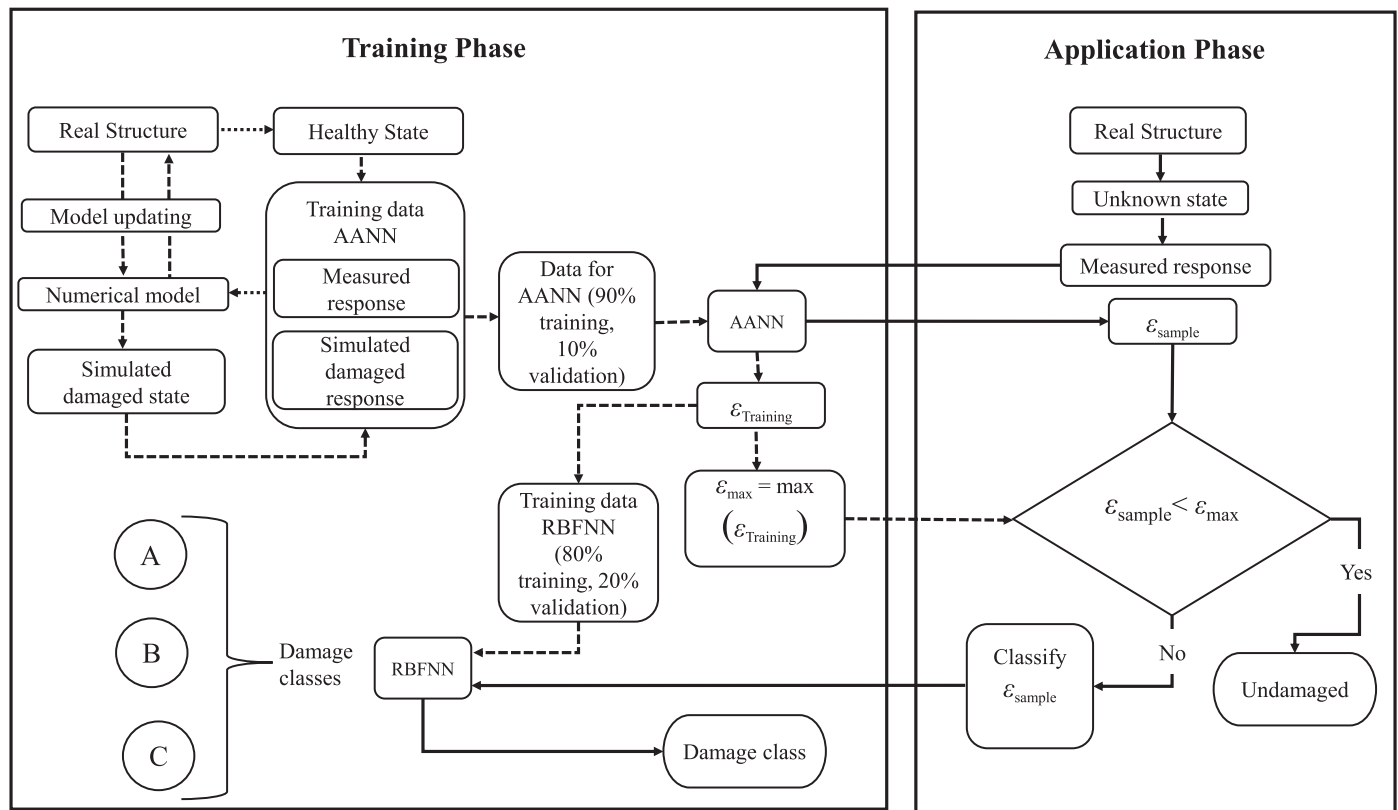


Fig. 4. Proposed two-step algorithm: dotted and straight lines signify the training and application phases, respectively.

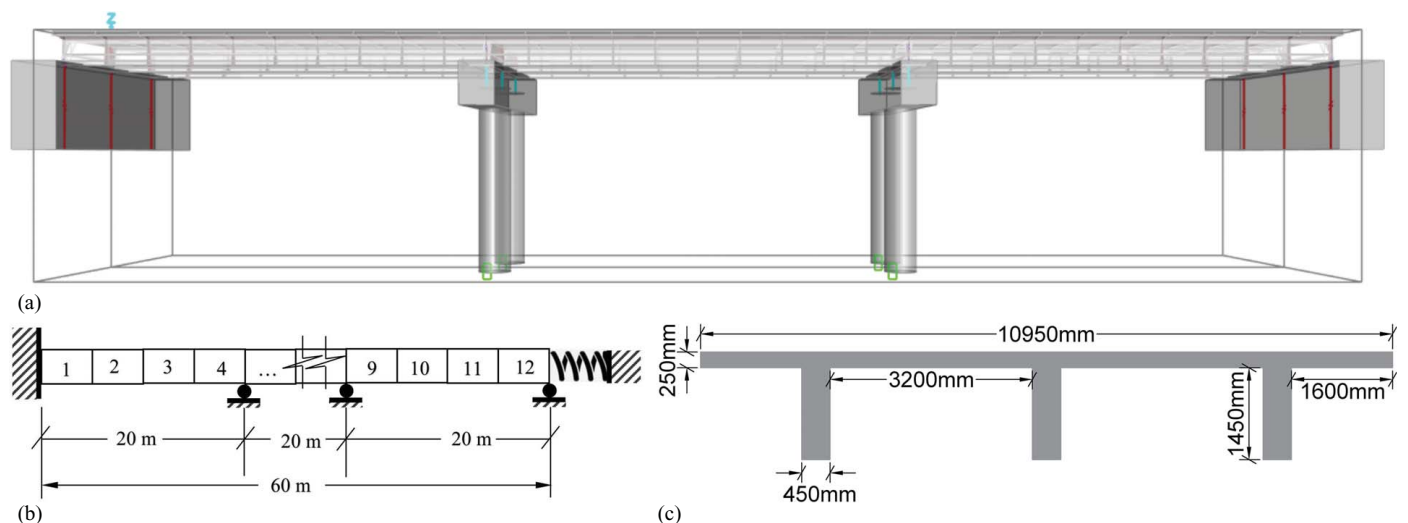


Fig. 5. Numerical bridge model: (a) numerical model; (b) beam idealization; and (c) bridge cross section.

and the assumed support conditions. The material properties are presented in Table 1. The elastic modulus, thermal coefficient, and Poisson's ratio were assumed to be temperature-dependent. The basis for the relevant interrelations was taken from Yan et al. (2005). The model was discretized using 12 equal-length two-noded Euler–Bernoulli beam elements with six degrees of freedom (three translational and three rotational) at each node. Each of these beam elements was also considered a segment of the bridge. Both the stiffness components (material and geometric), with temperature dependences demonstrated in the section “Temperature

Effect on Dynamic Properties of Structures,” were considered in the modeling.

The system was excited in all degrees of freedom using white Gaussian noise (WGN) as ambient forcing. The acceleration response was collected at vertical translational degrees of freedom at each node, with a sampling frequency of 256 Hz, which was later contaminated with WGN of different signal-to-noise ratio (SNR). The SNR is a measure of noise contamination level in which $X\%$ SNR signifies $X = 100\sigma(\text{noise})/\sigma(\text{signal})$, where σ = variance operation. The contaminated response signal was further

put through modal analysis, along with a peak-picking approach to identify the first five natural frequencies. The temperature data were assumed to be measured without any error. A total of 2,000 such experiments were performed on the undamaged condition of the structure under varying temperature. Next, damage was induced in the finite-element model through a percentage reduction of the modulus of elasticity (40%–60%) of the beam segments. For reference, an equivalent impact on stiffness could be created by reducing the elasticity for the entire length of the girder web by roughly 5% [as in Weinstein et al. (2018)] or by reducing the web depth for a single segment by around 10%–20%. Thus, the assumed measure for damage was consistent from its impact point of view with the damage measures commonly adopted in the literature (Weinstein

et al. 2018; Zhu et al. 2019). A total of 10,000 such experiments were performed under various damage conditions (location and severity) and the frequencies recorded. The identified frequencies for undamaged and damaged cases were further compared under different temperatures. It can clearly be seen from Fig. 6(a) that the bandwidth of the frequencies for both cases is almost of equal spread. Evidently, no conclusion can be drawn about the health of the structure simply by comparing the frequencies.

Prior to the validation study for the proposed two-step algorithm, the need for this algorithm was justified through a numerical experiment, in which the traditional ANN structure was employed to detect damage in the structures under temperature variation. For this, primarily, a four-layer feedforward network (Weinstein et al. 2018) was adopted with two hidden layers, temperature and frequency data in the input layer, and damage locations in the output layer. In the search for better precision, a third hidden layer was later added, forming a five-layer network. The ReLu function was used as the transfer function in the hidden layers, while the softmax function was employed for the final classification layer. The network, optimized for best possible performance, hosted 96, 128, and 96 perceptrons, receptively, in its hidden layers. No further improvement in the network was observed to enhance its

Table 1. Assumed material properties for the numerical model

Material property	Value	Geometric detail	Value
Density (kg/m^3)	2,400	Area (m^2)	4.69
E (GPa)	26.5	I_x (m^4)	2.79
Poisson's ratio	0.1	I_y (m^4)	27.382
G (GPa)	12.086	J (m^4)	30.18
Thermal coefficient ($^{\circ}\text{C}$)	1×10^{-5}	Temperature	-30°C to 60°C

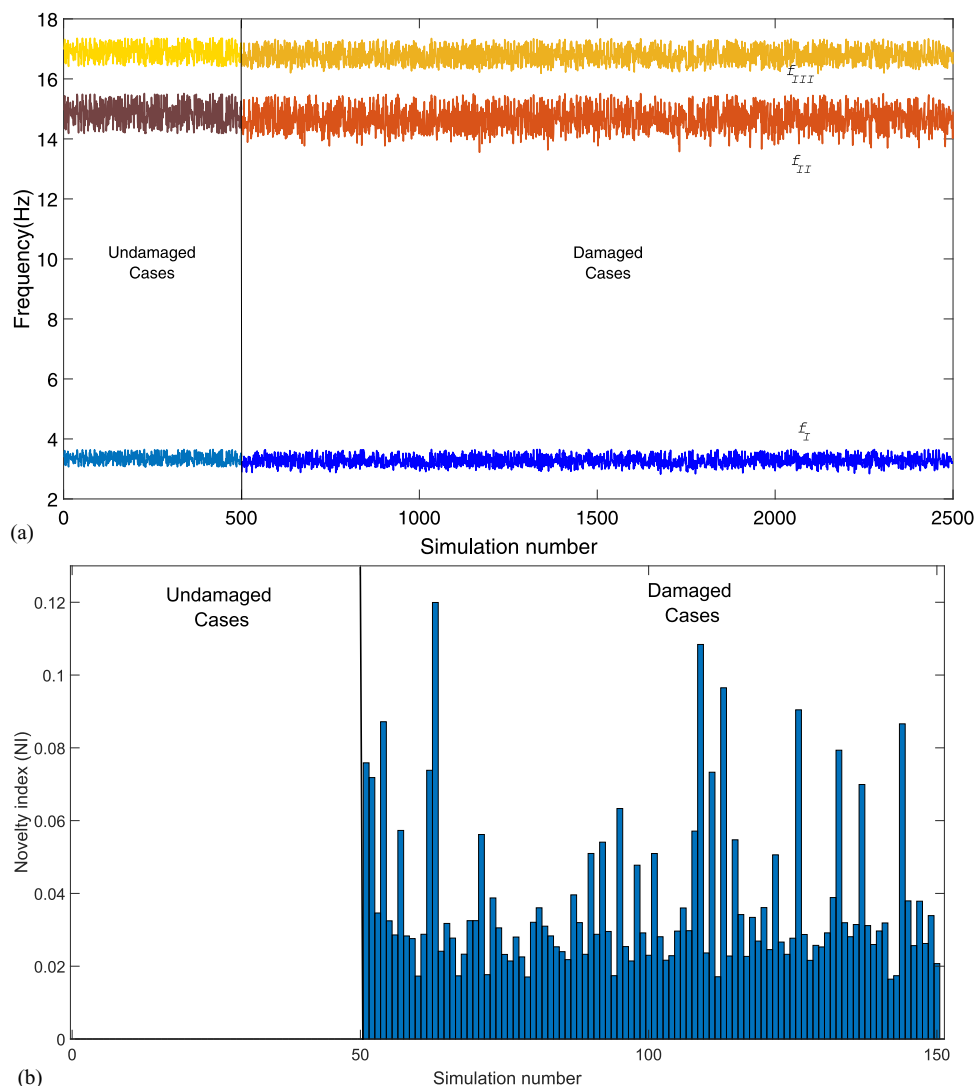


Fig. 6. Damage-induced changes in frequency and novelty index: (a) frequency variation under different temperatures; and (b) novelty index for damaged and undamaged cases.

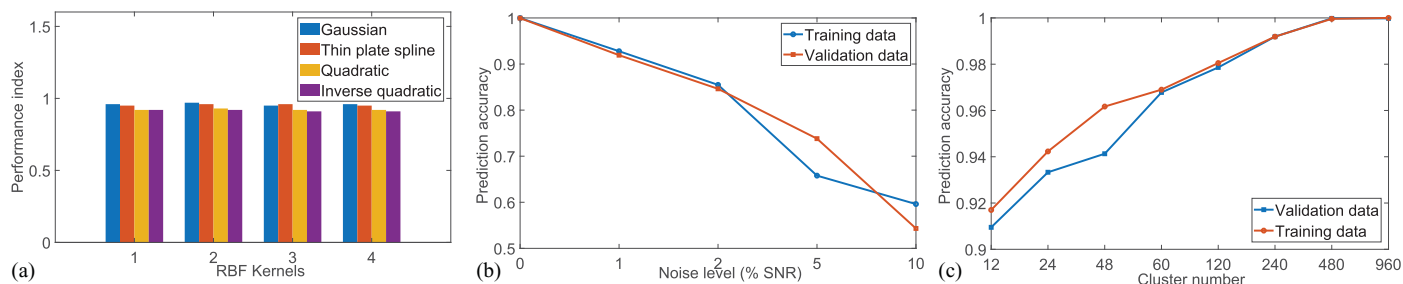


Fig. 7. Performance of classifier: (a) comparison of different RBF kernels; (b) performance with varying cluster number; and (c) performance of network with varying noise level.

accuracy significantly. With this, the best overall detection accuracies were achieved as 71.83% and 70.01% for the training and validation datasets, respectively. The average rates of true detection (D_{11}), true negative (D_{00}), false positive (D_{01}), and false negative (D_{10}) for all damage classes were obtained as 67.59%, 97.61%, 2.50%, and 35.11%, respectively. Evidently, for signals originating from structures under temperature variation, the traditional ANN architecture's performance in damage detection and localization is not satisfactory. The proposed algorithm was therefore tested for such problems to investigate whether a relative improvement could be achieved, demonstrated in the following.

First, 90% of the undamaged frequencies (1,800 samples), along with temperature, were used as the training input for the AANN to obtain the limiting value of the NI for classification as an undamaged dataset. The trained network was later validated using frequencies corresponding to 50 undamaged and 100 damaged cases. Fig. 6(b) shows that the damaged cases resulted in significantly large NIs compared with the undamaged counterpart. The accuracy of detection for this test case was found to be 100%.

Next, 80% of the entire frequency–temperature dataset (1,600 undamaged samples and 8,000 damaged samples) was put through the AANN to obtain corresponding prediction errors for classification with the RBFNN. The associated output heads were defined using a binary array of size equal to the number of beam segments (i.e., 12). Each of the entries in this array denoted whether or not the corresponding beam segment was damaged, with 0 and 1 denoting undamaged and damaged conditions, respectively. For example, the output head for damage at the seventh segment was denoted: $\mathcal{O}_7 = \{0 \ 0 \ 0 \ 0 \ 0 \ 0 \ 1 \ 0 \ 0 \ 0 \ 0 \ 0\}$. The RBFNN first clustered similar error patterns and subsequently associated them to particular damage types. To develop a multikernel approach, performance indices of four different kernels [Eq. (1)] were compared [Fig. 7(a)]. Quadratic and inverse quadratic kernels performed consistently poorer than the Gaussian and thin plate spline kernels for all trial datasets. The network was subsequently trained using a multikernel RBFNN approach with the Gaussian and thin plate spline kernels only.

Selection of the number of employed clusters was a major step in designing the RBFNN network. Large numbers of clusters may ensure better classification efficiency while reducing the computational economy. To maintain the balance, the optimized number of clusters was therefore decided using the elbow method. For this, the sum of squared errors (SSE) was observed for different cluster numbers to locate the position of a sharp shift in decreasing trend in SSE. It was found that the classification achieved sufficient accuracy with 480 clusters. The training and validation accuracies for increasing numbers of clusters are presented in Fig. 7(b).

Finally, the remaining 20% dataset (400 undamaged and 2,000 damaged samples) was employed to validate the trained network. The detection and localization accuracy was tested, employing several classifier performance assessment metrics, discussed next.

Performance Assessment of the Classifier

There exist several metrics to assess the reliability of any classifier. The confusion matrix (also called a contingency matrix), accuracy, sensitivity, and specificity are a few measures that are typically used for this purpose. This study follows the work of Tharwat (2020) to define a set of performance metrics for the proposed RBFNN classifier. The trained network's performance was investigated using confusion matrices for a multiclass classification problem. The confusion matrix is a table typically used to assess the performance of a classification model for a set of test data for which true values are known. For the present problem, it is basically a summary of the predictive ability of the network to locate damage. The confusion matrix shows the ways in which our classification model is confused when it makes predictions. The ratios of correct predictions over total test cases are summarized by diagonal elements of the normalized confusion matrix.

An elaborate set of simulations, involving 10,000 undamaged cases and 20,000 damaged cases, was additionally prepared. The classification results were further analyzed using the confusion matrix presented in Table 2. The results demonstrate that most of the damaged and undamaged cases were accurately localized and that instances of false detection were also significantly low. The diagonals in the confusion matrix signify true detection or true positive (TP). The false-positive (FP) and false-negative (FN) cases pertinent to a particular k th class signify the cases when the concerned segment was falsely identified as damaged or undamaged, respectively. The corresponding estimates can be obtained by summing up all the elements in the k th column or row except the k th element. True negative (TN) cases are those when the segment is truly identified as undamaged. All these metrics corresponding to the proposed RBFNN algorithm are presented in Table 3 for all damage classes (0: undamaged; N : damage in N th member).

The classifier's performance was also assessed for its accuracy, sensitivity, and specificity. Accuracy (ACC) is the ratio of instances of correct classification to the total number of cases (Sokolova et al. 2006). The sensitivity or true positive rate (TPR), or recall, is the ratio of accurately classified positive samples to the total number of samples. Finally, the specificity or true negative rate (TNR), or inverse recall, is the ratio of correctly classified negative samples to the total number of samples. All these metrics are defined in the following (Sokolova et al. 2006):

$$\begin{aligned} \text{ACC} &= \frac{\text{TP} + \text{TN}}{\text{TP} + \text{TN} + \text{FP} + \text{FN}}; & \text{TPR} &= \frac{\text{TP}}{\text{TP} + \text{FN}}; \\ \text{TNR} &= \frac{\text{TN}}{\text{FP} + \text{TN}} \end{aligned} \quad (8)$$

Additionally, Youden's index (YI) [also known as the bookmaker informedness (BM) metric] was obtained for the proposed classifier. YI combines sensitivity and specificity, and is defined

Table 2. Confusion matrix for the proposed classifier algorithm

	Target												
Prediction	0	1	2	3	4	5	6	7	8	9	10	11	12
1	0	93.09	4.07	0.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
2	0	6.41	93.77	0.18	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
3	0	0.37	0.55	99.58	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00
4	0	0.12	1.60	0.00	99.41	0.31	0.18	0.00	0.00	0.24	0.00	0.00	0.00
5	0	0.00	0.00	0.00	0.47	99.63	0.00	0.00	0.35	0.71	0.00	0.00	0.00
6	0	0.00	0.00	0.00	0.00	0.00	96.01	0.23	1.05	0.18	0.00	0.00	0.00
7	0	0.00	0.00	0.00	0.00	0.00	0.31	91.98	0.00	0.00	0.00	0.06	0.48
8	0	0.00	0.00	0.06	0.00	0.06	0.61	0.00	92.08	0.89	0.00	0.00	0.00
9	0	0.00	0.00	0.00	0.12	0.00	0.55	0.00	6.41	97.98	0.00	0.00	0.00
10	0	0.00	0.00	0.00	0.00	0.00	0.06	0.00	0.00	0.00	99.94	0.00	0.00
11	0	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	0.00	99.88	0.00
12	0	0.00	0.00	0.00	0.00	0.00	2.27	7.78	0.12	0.00	0.06	0.06	99.52

Table 3. Performance assessment of the RBFNN classifier

Class	TP	FN	TP2	FN2	FP	TN	ACC	TPR	TNR	YI
0	100.00	0.00	100.00	0.00	0.00	100.00	1.00	1.00	1.00	1.00
1	93.09	6.91	99.50	0.50	0.38	99.62	0.96	0.93	1.00	0.93
2	93.77	6.23	98.40	1.60	0.58	99.42	0.93	0.94	0.99	0.93
3	99.58	0.42	99.76	0.24	0.08	99.92	0.99	1.00	1.00	1.00
4	99.41	0.59	99.88	0.12	0.22	99.78	0.98	0.99	1.00	0.99
5	99.63	0.37	99.94	0.06	0.14	99.86	0.98	1.00	1.00	1.00
6	96.01	3.99	96.31	3.69	0.14	99.86	0.98	0.96	1.00	0.96
7	91.98	8.02	92.22	7.78	0.08	99.92	0.99	0.92	1.00	0.92
8	92.08	7.92	98.49	1.51	0.15	99.85	0.98	0.92	1.00	0.92
9	97.98	2.02	98.87	1.13	0.66	99.34	0.93	0.98	0.99	0.97
10	99.94	0.06	99.94	0.06	0.01	99.99	1.00	1.00	1.00	1.00
11	99.88	0.12	99.94	0.06	0.00	100.00	1.00	1.00	1.00	1.00
12	99.52	0.48	99.94	0.06	0.95	99.05	0.91	1.00	0.99	0.99

Note: ACC = accuracy; FN = false negative; FP = false positive; TN = true negative; TNR = true negative rate; TP = true positive; TPR = true positive rate; YI = Youden's index

as $YI = TPR + TNR - 1$. YI ranges from 0, signifying very poor performance of a classifier, to 1, for a perfect classifier. All these metrics (TPR, TNR, and YI) for the proposed classifier are presented in Table 3. Two other metrics (TP2 and FN2), additionally presented in Table 3, estimate whether the damage is detected or missed within one neighboring segment of the actually damaged segment, respectively. Table 3 demonstrates that the proposed algorithm raises significantly small false-positive or false-negative alarms. Moreover, if detection resolution is reduced to one in three segments, the false-negative alarm can further be reduced.

Next, the sensitivity of detection for different damage levels was tested for the proposed algorithm. For this, four different levels were selected for the induced damage: 10%–20%, 20%–40%, 40%–70%, and 70%–90%, as Levels 1, 2, 3, and 4. The objective of this experiment was to identify the practical damage intensity that can efficiently be detected with sufficient certainty with the proposed algorithm. For this experiment, datasets were simulated with 30,000 test cases (10,000 undamaged and 20,000 damaged cases) for each of the four damage levels, each of which was further contaminated with three different levels of noise (0%, 2%, and 5%), resulting in a total of 12 datasets of different damage intensities and noise levels. The proposed two-step algorithm for detection and localization was then employed for each dataset.

For all the case studies, it was observed that the first step (AANN-based occurrence detection) perfectly detected the presence of damage without fail. This established that the damage detection with the proposed algorithm was very much certain. The undamaged datasets were therefore removed and the RBFNN

classifier was employed to further locate the damage already detected in the first step. Substantial loss of performance was observed with increasing levels of noise, although it could be regained to some extent by employing larger numbers of clusters or selecting the clusters through higher numbers of iterations. Nevertheless, this approach increased the computational costs. For each of the case studies, cluster numbers and iterations were therefore varied until the maximum achievable accuracy was attained.

Case studies with uncontaminated responses (0% SNR) were observed to demand almost equal levels of computational effort, irrespective of damage levels. However, these cases were too idealistic and therefore not practical. With increasing noise levels, increased demand for computational efforts with performance deterioration was observed (Table 4). This clearly indicates that, for the practical application of the proposed damage detection algorithm, the limiting damage level depends on the noise levels as well. From this perspective, it can be concluded that Level 1 is too weak to be detected with certainty using this algorithm. It should be noted that, for Level 1, the algorithm attained an overall accuracy of 85% for noisy data, mainly owing to low FN cases, with FPs being significantly high. Similar to Level 1, the algorithm performed poorly in localizing Level 2 damages with sufficient certainty (Table 4). The associated FP, FN, and overall accuracy values were 12.86%, 1.17%, and 97% for 2% SNR and 18.57%, 1.69%, and 96% for 5% SNR. Thus, using this algorithm, Level 2 damage detection can still be advised, provided that the noise contamination is low or the desired degree of precision can be compromised. However, Levels 3 and 4, being capable of leaving

Table 4. Detection efficiency of the proposed algorithm for different damage and noise contamination levels

Damage (%)	Noise (% SNR)	Cluster no.	Iteration no.	Time (s)	FP (%)	FN (%)	ACC (%)
Level 1	0	300	350	830	0.00	0.00	100.00
	2	800	1,000	2,799	90.00	8.18	85.00
	5	1,000	1,500	3,984	90.00	8.18	85.00
Level 2	0	300	350	871	1.43	0.13	99.76
	2	700	800	2,317	12.86	1.17	97.00
	5	750	850	1,890	18.57	1.69	96.00
Level 3	0	300	350	564	4.29	0.39	99.29
	2	550	600	742	5.71	0.52	99.05
	5	550	700	1,805	2.86	0.26	99.52
Level 4	0	300	350	578	1.43	0.13	99.76
	2	500	600	1,791	4.29	0.39	99.29
	5	500	600	1,783	4.29	0.39	99.29

strong damage signatures in the signal, demonstrated sufficient robustness against noise contamination. The localization certainty was observed to be sufficiently high, while maintaining significantly low FP or FN values.

The proposed RBFNN was further tested under different noise contamination levels (0%, 1%, 2%, 5%, and 10% SNR) in collected response measurements. For this, the measured responses were contaminated with WGN sequences of specified SNR values prior to the modal analysis and peak-picking steps. The results are presented in Fig. 7(c). It can be observed that the algorithm's performance deteriorates very quickly with the noise contamination level; beyond 5% noise contamination, the prediction becomes very poor.

Discussion of Results

The proposed approach employs natural frequencies extracted from the measured data and employs them for damage detection and localization. It should be noted here that, unlike traditional modal information-based techniques, the isolated frequencies are not considered here as the damage-sensitive feature. Instead, the embedded correlation between several frequencies is used for damage detection. The AANN approach facilitates modeling this interrelation between frequencies and can track whether the same is missing in new data. This discrepancy is further attributed to the presence of damage. Further, the RBFNN uses the same interrelation information to classify different damage classes. Location-based damage indices are used in this article. Nevertheless, for real-life application, damage typology (deck delamination, girder corrosion, etc.) can also be used as damage classes. This nonmechanistic approach offers the flexibility to define the damage depending on the investigator's choice.

As it is a global parameter-based detection technique, the sensor location has less impact on detection accuracy. In strain gauge or acceleration time history-based techniques, the proximity of the sensor to the damage location is essential for damage to be detected with certainty; the present approach does not have such a condition. Also, the present approach does not demand simultaneous measurement or dense instrumentation; a single sensor can be sufficient to conduct the entire SHM process.

The proposed approach was successfully used to identify damage of different severities (reduction of 20%–40%, 40%–70%, and 70%–90% of undamaged elasticity of a single segment), except for a few cases that resulted in false predictions. It was found that the robustness of the algorithm against measurement noise is depleted

with weaker damage levels. However, the study identified that a 40% reduction in elasticity of a single segment, which can impart a change in the first natural frequency of less than 3%, can be detected and localized with sufficient certainty with the proposed algorithm. It was observed that a 40% reduction in elasticity in a single segment of the bridge creates an equivalent reduction in frequencies to a 5% reduction in elasticity for the entire girder span (as considered as damage in Weinstein et al. 2018) or a 20% reduction in girder depth at any location. Thus, the assumed damage severities can be considered to be realistic.

Damage detection using the AANN approach was found to be very efficient, with almost no false prediction. Thus, the false prediction obtained with the RBFNN can only be regarded as false location prediction, which is much less dangerous than a false-negative (type II) prediction. Thus, it is safe to conclude that the proposed approach always yields conservative projections of structural health.

Conclusion

A two-step neural network-based damage detection algorithm is proposed that can isolate the damage in a structure in the presence of temperature variations. An asymmetric AANN architecture with frequencies and temperature as input and frequencies as output is employed. The norm of the AANN prediction error, when employed as an NI, detects the occurrence of damage with high precision. Further, the prediction error is classified against the corresponding damage location using an RBFNN classifier. The proposed approach was tested using a numerical experiment on a multispan girder bridge and found to be prompt and precise in detection and localization, while sufficiently robust against temperature variations. Insignificant numbers of false-positive or false-negative alarms raised further demonstrated its reliability. The limiting value of noise contamination for which the algorithm can be applied was also identified.

The following was experienced:

1. Since the method is based only on frequency, the requirement of dense instrumentation or concurrent measurement is not compulsory; this further facilitates efficient bridge maintenance.
2. The occurrence detection accuracy for the proposed method is very high. Damage occurrence was detected for all case studies without fail, irrespective of damage and noise levels assumed.
3. The localization accuracy for such model-free data-based approaches significantly depends on noise contamination levels, as well as damage levels.
4. Detection accuracy can, however, be improved by recursive testing of the same structure using separate response data.
5. In the absence of real damaged response data, the proposed approach depends greatly on the accuracy of the supporting damage simulator model for damage localization.

Finally, based on the numerical experiment, it can be concluded that the proposed method holds the potential for real-life application in bridges, especially for those placed in subtropical regions, which experience significant temperature variation throughout the year.

Data Availability Statement

The following data that support the findings of this study are available from the corresponding author on reasonable request:

- model of the idealized bridge structure,
- pseudocode for the algorithm developed,
- training data simulation code.

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